Dam Structures and Water-Control Works Facility-Class Design Basis

The First Facility Class of the Dams Sector Design Basis

DBA-DAMS-DS-FC1 — Facility-Class Design Basis, DBA-DAMS Series

Issued Under: Dams Sector Framework (DBA-DAMS_Sector_Framework v0.3); DBA-MA Parent Framework

Version 0.4 — DRAFT

October 2026

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Working Draft — Released for Review and Discussion

SUGGESTED CITATION

T. Roxey, "DBA-DAMS-DS-FC1 — Dam Structures and Water-Control Works," Version 0.4, in CIKR Design Basis Examples, Edition 1 (November 2026), Eclectic Technologies, November 2026. https://doi.org/10.5281/zenodo.23863986

This version is a member of CIKR Design Basis Examples, Edition 1 (November 2026) (https://doi.org/10.5281/zenodo.23863986), a dated edition that freezes, for each of fifteen critical infrastructure sectors, a sector framework, a facility class and a reference event as they stood on 1 November 2026. The edition is not falsified by a later revision of this document, whose working master continues to move. It is a working draft, marked DRAFT on its cover.

1. Purpose and Scope

1.1 Purpose

This document sets out the design basis for dam structures and their water-control works, the first facility class in the DBA-DAMS series. The *Dams Sector Framework* (DBA-DAMS_Sector_Framework) states the sector's design basis in general terms, and this document applies it to one kind of facility. It names the events a dam and its works must withstand, traces how each event reaches the people below the dam and the facilities that depend on its reservoir, sets the parameters that size the dam's protection, and states what the dam must be able to do on its own when outside help is cut off. It describes no particular dam. A design basis for a specific dam, or one written for a state or a contract, starts from this document and adds only what is particular to it.

The framework calls this the highest-consequence class of the sector (framework §3.2). A dam holds back stored energy, and the failure of its structure or the misuse of its gates releases that energy downstream as a flood. The framework asks for an inundation consequence measure assessed for the deliberate case (framework §3.6, §9). On the author's decision of 12 October 2026 that measure is developed here, as a model that every instrument in the series can cite, and it lives in Section 4 of this document for now. Section 10.4 explains why, and what happens when it moves.

1.2 Scope

The class covers the dam and the works that control the water it holds. It includes the retaining structure itself, whether concrete gravity, arch, buttress or embankment, with its foundation and abutments. It includes the spillways, gated and ungated, and the auxiliary or emergency spillways and fuse plugs that back them up. It includes the gates and valves, with the hoists, drives, power supplies and controls that move them, and the outlet works that pass water through or around the dam. And it includes the instruments mounted on the structure that measure its condition: the piezometers, seepage weirs, survey points and accelerographs.

Three neighbors are separate classes of this series. The reservoir as a body of water, with its intakes, level control and appurtenant works, is *Reservoirs and Appurtenant Works* (DBA-DAMS-RM-FC2). Levees, floodwalls, hurricane barriers and tailings impoundments are DBA-DAMS-LV-FC3. The dam-safety programs, the Emergency Action Plans and the monitoring and warning systems that turn a measurement into a downstream warning are the sector's coordination class, DBA-DAMS-OpsCo. They are covered here only where the dam depends on them. One boundary inside the series is offered here for the framework to decide: the instruments on the structure are this class's hardware, and the detection and warning function they feed is OpsCo's (Section 10.6).

Three boundaries with other series are drawn on the rule the framework states, that the served function is the partner series' and the structure and its release are this series' (framework §6.3).

Hydroelectric generation is *Large Hydroelectric Storage* (DBA-ES-HY-FC1). Water-supply storage is *Storage Reservoir* (DBA-W-FC3). Navigation locks are *Ways and Linehaul* (DBA-TRAN-WY-FC2), which on the author's decision of 8 October 2026 takes the lock chamber, its gates and valves and their machinery, and leaves the dam, the impoundment and their release to this series. Where a water passage serves a partner function, such as a penstock or an intake, the release of the impoundment through it is this class's and the loss of the served function is the partner's. Section 8.3 states the boundaries, and Section 10.3 records how the boundary with the hydroelectric class was agreed with Project 8 and carried out.

What makes a structure a member of this class is the same at every site: it holds back a mass of water whose sudden release would flood the land below, and it carries the works by which that water is released on purpose. The particular structure, river and downstream valley differ from dam to dam, and that detail belongs in each dam's own design basis, not here.

1.3 The Reference Structures

Three kinds of structure are used as reference cases. Each was chosen because it shows a different way a dam can end up releasing its reservoir.

The high-hazard concrete gravity dam with a hydroelectric plant and a gated spillway holds its reservoir by the weight of its concrete and passes floods through large gates set in its crest. Concrete does not wash away the way an embankment does, so its accidental failure is rare, but when it is breached it opens quickly and fully, and the warning time downstream is short. It is the governing configuration for the deliberate breach and for the misuse of the gates, because its gates can release a large flood on command and its powerhouse ties it to the electric series. It is the structure the framework names in its reference event, the destruction of the Kakhovka dam on the Dnipro in June 2023 (framework §1.5).

The large embankment dam with a gated spillway, serving water supply, holds its reservoir behind a mass of compacted earth or rock and depends on its spillways to keep water from flowing over the top. An embankment that is overtopped erodes and fails. Its failure usually develops over hours, not seconds, so warning is possible, but the volume behind it is large and the decisions taken while it develops decide the outcome. It bounds the hydraulic path: the failure of a spillway, the choice between damaging one release path and opening another, and the evacuation that follows. The Oroville spillway incident of February 2017 is its case. Because the reservoir supplies water as well as generating power, the loss of the pool reaches the water series.

The cascade of embankment dams on one river is a chain of dams in which each reservoir sits on the one below. It bounds the propagation path: when one dam fails, its flood arrives at the next, which was not designed to receive it, so a single failure becomes several. Cascades of old, privately owned dams with modest spillways are common, and responsibility for them is often split between regulators. The failures of the Edenville and Sanford dams in Michigan in May 2020 are its case.

A requirement that names one of the three structures applies to that kind of structure. A requirement that names none applies to all three. Any other dam follows a requirement to the extent it shares the structure, works or setting the requirement addresses. A fourth event, the intrusion into the control system of the Bowman Avenue flood-control dam in New York in 2013, is not a reference structure. It is small, but it characterizes the path by which a gate is reached through a network, and that path scales to every structure in the class (Section 5.5).

1.4 Class Type

This is a physical facility class under the author's determination of 30 September 2026 (*Facility-Class Types Are Physical, Coordination and Logical*). The harm it can do comes from the structure and the water the structure holds, rather than from the people who coordinate it or from software services. In the framework's structure it sits under Dam Structures and Water-Control Works (DS), the first of the sector's three function classes (framework §3.1, §3.2).

1.5 Relationship to Governing Documents

This instrument is issued under the *Dams Sector Framework* and inherits the DBA-MA parent framework. It does not repeat the framework's taxonomy, bounding principles, consequence measure or Bright Line and Command Broker architecture; it inherits them, and Section 9 is the required inheritance map. Where the framework is still a skeleton (framework §4, §5 and §7), this instrument develops the class-level content ahead of it and records in Section 9 what the framework should take up. Section references of the form "framework §N" resolve against the *Dams Sector Framework* in force.

Governing instruments. This instrument does not own the concepts it applies. The Bright Line — its placement criterion, its granularity, its scope of application and the obligation it creates — is defined in FD-BL (*The Bright Line*), which governs. The Broker role that discharges that obligation is defined in FD-BR (*The Broker*). Layer decomposition, the determinism boundary as a scope gate, and the machine-learning carve-out are defined in FD-LD (*Layer Decomposition and Determinism*). Local sovereignty, the property of a protective function that executes independently of reachable digital authority, is defined in FD-LS (*Local Sovereignty*). Two determinations bind under FD-BL: FD-BL-D1 (*Mode of Discharge*) governs how the authorization obligation may be discharged, and FD-BL-D2 (*Indication Integrity*) governs the integrity of the basis on which an authorization is granted. FD-EV (*Envelope*) reserves the term "envelope". Placement is on the consequence of the individual action, assessed per action; this instrument applies that definition and does not vary it. The Foundational Definitions are cited without a version token, and FD-REG (*Governing Stack Register*) is authoritative for their version state.

2. The Dam Structures and Water-Control Works Facility Class

2.1 Defining Characteristics

For design-basis purposes, the dam is defined by three features. First, it holds stored energy. The water behind it has weight and height, and the higher the pool stands, the more energy is held and the larger and faster the flood its release would make. Second, that energy can leave by two routes. It can leave through a breach of the structure, which no one intends, or through the works, which exist to release water on purpose. A gate that opens at the wrong moment, or fails to open when it must, can do much of what a breach does. Third, the dam's protection depends on decisions taken in time: when to release, how much, through which path, and when to warn the people below. The gates, their controls and the instruments and forecasts on which those decisions rest are where a cyber attack would aim, so they are the focus of the cyber requirements.

Stored energy, two routes for its release, and decisions that must be taken in time: these three define the class. They set it apart from the reservoir class, which manages the stored mass without being the barrier, and from the levee class, whose structures hold back water only when a flood arrives.

2.2 The Hazard-Potential Sub-Index

The class is sub-indexed by hazard potential under the framework's anchoring (framework §2.2, §3.1). The federal guidelines rate a dam by what its failure or mis-operation would do downstream, not by its condition. A high hazard potential dam is one whose failure or mis-operation will probably cause loss of human life. A significant hazard potential dam is one whose failure or mis-operation would probably cause no loss of life but could cause economic loss, environmental damage or the disruption of lifeline facilities. A low hazard potential dam is one whose failure would probably cause no loss of life and limited losses, mostly to the owner's property (*Federal Guidelines for Dam Safety: Hazard Potential Classification System for Dams*, FEMA P-333). The guidelines name mis-operation beside failure, which is the second route of Section 2.1.

All three reference structures are of the kind the high hazard rating describes. The classification is assessed today for an accidental failure, on a clear day or in a flood. Section 4 adds what the classification does not assume: a release at the pool level, the hour and the coincident conditions an adversary would choose, with the warning denied. A dam rated significant or low for an accidental failure may rate higher on that basis, and the class records the deliberate-case rating beside the accidental one rather than replacing it (Section 4.5).

2.3 Deployment Typologies and Size Range

The class spans federal, state, utility and private owners. FEMA's page for the National Inventory of Dams gives 92,075 dams in the inventory, and the American Society of Civil

Engineers' 2025 report card counts more than 16,700 of them as high hazard potential. Many of the high-hazard dams are small, old and privately owned, and most are regulated by the states rather than by a federal agency (framework §2.1).

Dams also form cascades, several on one river with each reservoir backed against the dam above. A cascade raises the question the refinery class asks of a cluster: what happens if several are lost together (*Petroleum Refining* §2.3). For a cascade the answer is not a coincidence that an adversary arranges; the river arranges it, because the failure of one dam sends its reservoir to the next. The class applies across the size range. A small dam above a town can do more harm than a large one in an empty valley, and what counts in this framework is the severity of the harm, not the size of the dam (framework §4.1).

2.4 Consequence Context

The dam's governing consequence is the flood its release sends downstream. The framework measures it as a peak, not a duration: the inundation zone and the population at risk within it (framework §3.6). This class resolves that measure into the depth and speed of the water where people are and the time between the onset of the release and the arrival of the flood (Section 4). Its time-to-harm is set by the travel of the flood wave, which can reach the nearest population in minutes, and not by any return-to-service schedule.

The dam has a second kind of harm, slower and wider, that the framework names as the multi-purpose dependents (framework §6.2). When the structure fails, the reservoir is lost as well as released. Everything that drew on the pool loses it at once: the generation the electric series counts, the supply the water series stores, the navigation the transportation series carries, the irrigation that farms rely on, and the cooling water of any plant that takes it from the reservoir. A reservoir does not come back when the structure is repaired; it has to refill, and the structure has to be rebuilt first. That recovery is the framework's secondary, long-lead consideration (framework §3.6).

Kakhovka shows both kinds of harm from one act. On 6 June 2023 the dam was breached and its reservoir, one of the largest in Europe, began to drain. The joint assessment by the Government of Ukraine and the United Nations of 17 October 2023 counted about 620 square kilometers submerged across four regions and about 100,000 residents directly affected. Upstream, the Zaporizhzhia nuclear plant drew its cooling water from the same reservoir. The International Atomic Energy Agency reported that day that if the level fell below 12.7 meters the plant would no longer be able to pump from it, and that its cooling pond held enough water to supply the plant for several months, its six reactors being in shutdown. The flood went downstream; the loss of the pool went upstream. The lesson the class takes from Kakhovka is not the means but the shape of the event: a single act opened both consequences, at a moment chosen by whoever acted.

3. The Facility-Class Design Envelope

3.1 Why Dam Structures Are the First Class

The framework calls the dam-structure class the highest-consequence class of the sector (framework §3.2). This document takes that ordering as given. The range of events the dam must withstand is set by the two ways its failure reaches people, described in Section 3.2.

3.2 The Two Consequence Chains

Chain I — inundation. Overtopping, structural or foundation failure, failure of a spillway, mis-operation of the gates, or deliberate action releases the stored water uncontrolled. The chain runs through the river valley to the people and facilities below the dam, on the timescale of the flood wave. It is the governing chain, and its consequence is set by the inundation model of Section 4. In a cascade it continues to the next dam down, which becomes part of the release.

Chain P — loss of the pool. The impoundment is lost or drawn down below the level on which a dependent facility relies, for longer than that facility can manage without it. The chain runs through the dependents of the reservoir, upstream as well as down: the intake of a water system, the turbines of the dam's own plant, the irrigation canal, the cooling water of a power plant on the shore. It activates on the timescale of the dependent's own reserve, which may be hours for a canal and months for a nuclear plant's cooling pond. It reaches the dam whether the pool is lost by breach, by an uncontrolled release through the works, or by a deliberate drawdown taken to protect the structure.

A breach opens both chains from one act: it floods the valley and empties the reservoir. A protective drawdown opens only Chain P, at a cost the dependents bear so that Chain I cannot open. That trade is a decision, and Section 6.7 asks who takes it. The design basis carries both chains.

3.3 Design Parameters at Class Resolution

Seven parameters govern the envelope. The first is the governing release: the volume and head of water a single credible event can release, taken at the pool level the event assumes. For the deliberate case that is the highest level the dam is permitted to hold, because an adversary chooses the moment, and at Kakhovka the reservoir stood high in the weeks before the breach, as reported. The second is the inundation footprint: the area, depth and speed of flooding and the population within it, under the conditions Section 4 specifies. The third is the warning time: the time between the first detectable sign of a release and the arrival of the flood at the nearest population. It is the parameter that decides how many people the flood finds, and Section 4.5 states how it is counted.

The fourth is the release capacity: the discharge the works can pass with the gates actually available, against the inflow design flood. It is stated for the gates as they can be operated, not as drawn. Edenville was found by the Federal Energy Regulatory Commission in 2018 able to pass

about half of its probable maximum flood, and that figure assumed gates that could be opened fully, which in May 2020 they were not, on press accounts of the Independent Forensic Team's final report. The fifth is the decision margin: the time from a credible forecast or first indication of trouble to the point at which a release decision forecloses the others, within which the pool is lowered, a damaged path is used or abandoned, and the downstream population is warned, with the trigger for each stated in advance. At Oroville the Independent Forensic Team found that decisions were taken to limit the flow through the damaged spillway so as to keep water off the emergency spillway, and that once the emergency spillway was allowed to overtop, the risk it had carried was soon realized.

The sixth is the cascade loading: for each dam in a cascade, the flood that the failure of every dam above it would deliver, against what the dam can pass or survive. The owner's own incident report on Edenville, as the Independent Forensic Team's interim report quotes it, says that it was always understood by the regulators and engineers involved that should a breach occur at Edenville, Sanford would necessarily fail. The seventh is the pool-dependency margin: for each facility that depends on the reservoir, the time it can continue after the pool falls below the level it needs. It governs Chain P, and the dependent's reserve, not the dam's, sets it.

This instrument states the parameters and their governing logic. Their quantitative values await validation against the dam-safety analyses that owners already hold: the inundation maps of their Emergency Action Plans, their potential failure modes analyses and their spillway adequacy studies (Section 10.1). They are not asserted here as numbers.

4. The Shared Inundation Consequence Model

4.1 Status and Use

The framework requires an inundation consequence measure for the deliberate-breach case (framework §3.6, §9). On the author's decision of 12 October 2026 this section is the model's provisional home, as the chemical-release model is held in *Chemical Production and Synthesis* §4, and it is held here until the framework confers a designation on a standalone instrument. An instrument citing the model cites this section, by title and section. When the model moves, the citing instruments are owed notice that a sweep is due (Section 10.4).

The model has four stages: release, flood routing, endpoint and exposure. It states what each stage must account for. It does not supply a breach or routing code, and it does not displace the inundation analyses that owners already perform for their Emergency Action Plans. Those analyses follow federal guidance that asks for a fair-weather breach from the normal pool, often called a sunny-day failure, and a breach during the inflow design flood, which for a high-hazard dam is typically the probable maximum flood (FEMA P-946). The model takes both as its floor and adds the deliberate case, setting the conditions an owner's analysis must meet to serve as a design basis.

4.2 Release

The release is the volume, rate and duration of water leaving the reservoir. For the design basis it is taken from the governing release of Section 3.3, and four conditions are added to what an accidental-failure analysis would assume. First, the pool stands at the highest level the dam is permitted to hold, not at its normal level, because a deliberate act chooses its moment. Second, the breach forms as fast as the means used can make it. An embankment eroded by overtopping opens over hours; a breach made by explosives opens at once, and a concrete section lost to a strike goes in one piece. The rate of breach formation assumed for an accident is not credited against deliberate action. Third, more than one breach is possible. A deliberate act can open the structure at more than one place, and the release is counted for the openings it could make together.

Fourth, in a cascade the release is the sum of what fails. The flood from the dam above arrives at the dam below with its own reservoir behind it, and if the lower dam fails or is overtopped, its reservoir joins the flood. The release is counted for the cascade the failure can reach, not for the single dam where it began. At Edenville, on the interim report of the Independent Forensic Team, the embankment failed at about 5:35 in the evening of 19 May 2020, and its flood reached Sanford, which was overtopped about two hours later and breached. Mitigation that depends on the works the event itself defeats is not credited. A gate that needs grid power is not credited against an event that takes the grid, and a gate that can be reached through a network is not credited against an attack on that network.

4.3 Flood Routing

Routing carries the release down the valley and gives, at each place that matters, the depth and speed of the water and the times at which it arrives and peaks. Federal guidance asks the inundation map to show the areas that would be flooded and the travel times of the wave front and the flood peak at critical locations (FEMA P-64). The model takes that as its output, with three additions. The routing is done for the release of Section 4.2, not only for the owner’s two standard cases. It carries the flood through every structure in its path, counting a bridge, a levee or a downstream dam as something the flood can remove as well as be slowed by. And it states the state of the river when the release arrives: a breach timed to a flood in progress puts its water on top of water already out of the banks, and an adversary who can choose would choose that.

4.4 Endpoint

The endpoint is the condition of the water at which a person caught in it is likely to be killed or seriously harmed: a combination of depth and speed in which people cannot stand, vehicles are carried away, or buildings fail. The design basis uses the endpoint at which people can no longer save themselves by moving to higher ground, because that is the protective action the Emergency Action Plan depends on. Its values are taken from the dam-safety practice the owner already follows and are not set here (Section 10.1). The arrival of the flood at a place is counted from the first rise of the water, which FEMA’s flood risk guidance for dams takes as a one-foot increase

in water-surface elevation (Guidance Document 68), and not from the arrival of the endpoint, because the first rise is when a warning is overtaken.

4.5 Exposure

Exposure is the population within the footprint and the time they have to act. It accounts for who is present by hour, for places whose occupants cannot easily move, such as hospitals, schools, care homes and jails, and for the facilities of other sectors inside the footprint. It counts the warning time of Section 3.3 at each place, and it counts that time from the first detectable sign of the release, not from the moment the owner declares an emergency. Under a positive applicability determination, two anomalies in the dam's own indication are such signs. The first is the loss together of the dam's detection and its warning paths, because an adversary who means to breach the dam silences it first. The second is a divergence between the independent indication of pool level and gate position and the indication of the system that commands the gates, which Sections 5.4 and 5.5 require to be alarmed. The warning time is counted from either, and the valley is warned on it, not only on the flood. Federal guidance observes that a sunny-day failure generally carries the most potential for loss of life because of the element of surprise (FEMA P-64). A deliberate breach is a sunny-day failure that an adversary has chosen, and it may come with the warning denied. The model therefore states the exposure twice: once with the warning the Emergency Action Plan provides, and once with the warning that survives the denial the event carries (Section 6.2). The difference between the two is what the warning system is worth, and it is stated, not assumed.

Exposure also counts what the flood reaches without choosing it. It counts the facilities in the footprint whose own hazards the water can release: a chemical plant whose ponds and tanks the flood can enter, a water treatment plant, a substation. During the Edenville and Sanford floods the Dow chemical plant at Midland reported floodwater commingling with its on-site containment ponds; the company said there was no threat, and the state's preliminary report found no material impact on the river system. It counts the routes by which people leave and responders arrive, and the bridges the flood may take. And it counts the dependents of the pool under Chain P, upstream as well as down, with the time each can last. The model's output for any instrument citing it is the footprint, the population at risk, the warning time at each place, the facilities within the footprint, and the dependents of the pool with their margins.

The class records a deliberate-case hazard rating beside the accidental one (Section 2.2). A dam whose deliberate-case footprint reaches a population its accidental-case footprint does not, because the pool stands higher, the breach opens faster, the river is already in flood or the warning is denied, carries the higher rating for the purposes of this design basis. The owner's regulatory rating is unchanged; the design basis records the difference.

5. Design Basis Events: Non-Adversarial Envelope

The framework lists candidate bounding events for the sector but does not yet develop them (framework §4.1). This section develops each for the dam structure, using the series' four headings: the event, the harm that makes it count, what the dam must do, and for how long. Durations are stated as class parameters awaiting the framework's values (Section 10.2).

5.1 Hydrologic Loading Beyond Release Capacity

Postulated event an inflow larger than the works can pass at the gates available, raising the pool toward the crest of the dam, whether from rainfall, snowmelt or the failure of a dam upstream; or waves on the pool, raised by wind or by a landslide into the reservoir, that run up beyond the freeboard the dam allows.

Consequence threshold water over the crest of an embankment, or loading on a concrete structure beyond its design, threatening a breach whose footprint reaches a population (Chain I).

Required function the works must pass the inflow design flood with the gates as they can actually be operated, and the dam must state the margin between that flood and the flood that would overtop it. Where a gate cannot be opened fully, its limit is counted in the release capacity of Section 3.3. Where the dam cannot pass the inflow design flood, the shortfall is a departure from the design basis and not an accepted condition, and the protective measures that stand in for it, a lowered pool or a staged evacuation trigger, are stated in advance with their triggers. The freeboard is stated for the design-basis wind acting on the pool at the highest level the dam is permitted to hold, and, where the slopes of the reservoir can produce one, for a wave raised by a landslide into it (Section 5.8). A wave that runs over the crest of an embankment is counted as overtopping, however brief. The Federal Energy Regulatory Commission revoked Edenville's license in September 2018 for the owner's long-standing failure to increase the project's spillway capacity to pass flood flows safely; twenty months later the dam failed in a flood that the Independent Forensic Team's chair put at between a 100-year and a 200-year flood, as reported. Discharge through the penstocks and turbines of a hydroelectric plant at the dam is a release of the impoundment and is this class's (Section 8.3). It may be counted in the release capacity of Section 3.3 against the events of this section only while the wicket gates and the hydraulic systems that move them are shown intact and operable and the units can pass the flow. Its availability depends on the generating plant, so it shares a cause with the generating chain: it cannot be assumed after a strike on the transformer or the generator, which may come in the same campaign that raises the need for release. It is therefore not credited against deliberate attack (Section 5.6), and it is not counted in the release capacity on which the Autonomous Safe Impoundment Condition rests (Section 6.7). The condition on which it depends is stated in *Large Hydroelectric Storage* §7.4 and is cited here, not restated.

Design-basis duration the duration of the inflow design flood and its recession; until the pool is back within its normal band.

5.2 Structural or Foundation Failure

Postulated event loss of strength or stability in the structure, its foundation or its abutments: sliding, cracking, internal erosion of an embankment or its foundation, or the sudden loss of strength of a saturated embankment, whether under normal loading or under the high pool of a flood.

Consequence threshold a breach, or a condition that will become one, whose footprint reaches a population (Chain I).

Required function the structure must be shown stable along its whole length under the loads it can carry, and not only at the sections that were analyzed. Every potential failure mode the owner has identified carries a means of detecting it in time to act, and the margin between detection and breach is stated. At Edenville, on the interim report of the Independent Forensic Team, the most plausible principal mechanism of failure, with strong evidence, was the static liquefaction of saturated, loose sands in the downstream section of the embankment; overtopping was ruled out. A structure can fail in a flood it can pass. The class therefore requires that the stability of the structure at a high pool be part of the design basis for every flood the dam is expected to pass, and not only its hydraulic capacity.

Design-basis duration event-specific; until the structure is shown stable or the pool is lowered to a level at which it is.

5.3 Failure of a Spillway or Outlet Works

Postulated event damage to a spillway chute, its foundation or its energy dissipation while it is passing flow, or erosion below an emergency spillway when it is used for the first time, or failure of an outlet conduit.

Consequence threshold loss of a release path while the pool is high, forcing the use of another path that may itself fail, or erosion that advances toward the control structure and threatens an uncontrolled release (Chain I).

Required function every release path the dam relies on to pass the inflow design flood must be shown able to carry its design flow for the duration of the flood without damage that would cut it off, including a path that has never been used. Where the dam relies on an emergency or auxiliary spillway, the ground below it is part of that spillway. The decision between using a damaged path and opening another is planned in advance, with the conditions under which each is chosen and the downstream warning that each requires. At Oroville, on the Independent Forensic Team’s account, the service spillway chute slab cracked and was lifted by water entering through cracks and joints over poor foundation in places, and when flow was held down to keep water off the untested emergency spillway, the emergency spillway was overtopped and eroded, and the evacuation order became a necessary precaution.

Design-basis duration the duration of the flood the path must pass; until the pool is lowered below the path’s crest.

This category is characterized empirically by the Oroville spillway incident of February 2017, conferred by the framework as DBA-DAMS-RE-2 (framework §9.1). On 7 February 2017, with releases being raised, the operators of the California Department of Water Resources saw an unusual flow pattern on the service spillway, stopped the flow and found the chute eroded. Releases were resumed and limited while the reservoir rose. On 11 February water flowed over the emergency spillway for the first time in the facility's history, and erosion below it began to advance toward the weir. On 12 February the Butte County Sheriff ordered evacuation, and some 188,000 people downstream were ordered to leave, as reported at the time. The Team found that the incident was caused by a long-term systemic failure of the Department, regulatory and general industry practices to recognize and address inherent spillway design and construction weaknesses, poor bedrock quality and deteriorated chute conditions, and that compliance with regulatory requirements is not sufficient to manage risk. The design-basis reading is that a release path the design relies on but has never used is a hypothesis, not a credit, and that the decisions taken while a dam is in trouble belong in the design basis, made in advance.

5.4 Loss of Gate Power, Gate Function or Control

Postulated event loss of the power that moves the gates, or of the power that runs the instruments on the structure, beyond the capacity of on-site backup; mechanical failure of a gate or hoist; or an erroneous command from the control system that opens, closes or holds a gate against the condition of the reservoir.

Consequence threshold a gate that cannot be opened when the inflow requires it, raising the pool toward overtopping; or a gate opened when it should not be, releasing a flood downstream without warning (Chain I); or an erroneous release that draws the pool below the level its dependents need (Chain P).

Required function every gate the dam relies on for its release capacity must be operable, at the opening the design basis credits, without grid power and without the networked control system, by a means that does not share the cause that removed them: backup power protected against the same flood, and manual or local operation by people who can reach the gate. The instruments on which the detection of a potential failure mode rests (Section 5.2) carry backup power for the design-basis duration, or a means of manual observation stated in advance with the people who make it, so that the loss of the grid does not also blind the dam; an instrument with neither is not credited in that detection. No gate movement that could release a flood reaching the endpoint downstream may be commanded by any system an attacker could reach except through a deliberate, authenticated and recorded act of a qualified person that no automated system can perform or finish alone. FD-LS (*Local Sovereignty*) calls this a governed act, and the term is used in that sense throughout. The position of every gate is indicated to the operator by a means independent of the system that commands it, and that indication stands continuously beside the indication of the commanding system and is compared with it, with a divergence between the two alarmed (Section 5.5). Where a gate's opening is limited for its own safety, as Edenville's gates were held below full opening in May 2020, on press accounts of the Independent Forensic

Team's final report, the limit is a reduction in release capacity and is counted under Section 5.1. Every gate whose release can reach the endpoint downstream has the rate at which it opens limited by its machinery, so that below a pool and river condition stated in the design basis no command, authorized or compromised, can raise the river at the nearest population faster than the warning time of Section 3.3 allows. Above that condition, where the release capacity of Section 5.1 or the fast opening of a lower dam's gates ahead of an arriving flood (Section 5.7) requires it, the gates open at the rate the release needs. The condition, and the rate on either side of it, are stated in advance. The change from the limited rate to the full rate is made at the structure, by the measured condition of the pool and river, and no system an attacker could reach can make it except through a governed act. The limit is set in the machinery, not in the control system, and it is one of the protective functions of Section 7.2.

Design-basis duration the duration of the inflow event for the failure to open; event-specific for the erroneous release.

5.5 Cyber Attack on Gate Control and Dam-Safety Monitoring

Postulated event intrusion into the control system of the gates or into the monitoring system of the structure, which reads the state of the dam, alters the indication presented to operators, or commands a gate.

Consequence threshold a gate commanded to open at a high pool, or held closed during a flood, while the operators see normal readings; or a developing structural failure hidden from the people who would warn the valley.

Required function the gate controls must be reachable for a change of position only through a governed act (Section 5.4), and the controls and the monitoring system must not be reachable from any network an attacker could reach without that act. Every remote access path to a gate controller, including a cellular modem or a vendor's maintenance connection, is part of the design basis and is recorded, and one not recorded is a departure. The record is verified against the controllers themselves, by inspection at the gate and the control house and not only from the owner's documentation, at an interval the design basis states, because a path added after the record was made is the path an intruder is likely to use. For a dam the Federal Energy Regulatory Commission licenses, the verification belongs in the comprehensive assessment its regulations require (Section 8.4); for a dam outside federal jurisdiction it falls into the gap recorded at Section 10.8. The readings on which an operator decides whether to release, or whether to warn, must be checkable by a means that does not depend on the system that may have been compromised. The check is continuous: the independent readings stand beside the operators' readings and are compared, and a divergence is alarmed, because a clean falsification shows no anomaly of its own and the class does not rely on an operator's choice to look (Section 6.4). A gate that cannot be moved because it happens to be disconnected for maintenance is not credited against attack.

Design-basis duration indefinite; the condition persists until the compromise is found and removed.

This category is characterized empirically by the intrusion into the Bowman Avenue Dam in Rye Brook, New York, conferred by the framework as DBA-DAMS-RE-4 (framework §9.1). On the Justice Department's account of 24 March 2016, between 28 August and 18 September 2013 the charged individual repeatedly obtained unauthorized access to the dam's supervisory control and data acquisition system and obtained information about its water levels, temperature and the status of its sluice gate, which controls water levels and flow rates. That access would normally have permitted the remote operation of the gate, but the gate had been manually disconnected for maintenance at the time. The United States Government stated that the defendants worked for two Iran-based computer companies that performed work on behalf of the Iranian Government, including the Islamic Revolutionary Guard Corps. The dam is a small flood-control structure about twenty feet high, on the Associated Press's account, and access was reportedly gained through a cellular modem, on press accounts of an unclassified Department of Homeland Security summary. Investigators first believed, as reported, that the target was a much larger dam of the same name in Oregon. The design-basis reading is that a gate reached through a network was stopped by a maintenance outage, not by the design, as at TRITON, where the safety controllers' diagnostics turned the attacker's own error into a safe state (*TRITON Attack* F-4), and that the design basis cannot count on the next intrusion finding the gate disconnected.

5.6 Physical Attack on Critical Nodes

Postulated event a physical attack on a single node whose loss defeats the release capacity or the warning: the gate hoists and their drives, the gate control house, the power supply to the gates, the monitoring cabinets, or the communications path that carries the alarm.

Consequence threshold the loss of release capacity during an inflow, or the loss of the warning, across more than one gate or path.

Required function such nodes must be hardened. Where release capacity or warning is provided by more than one path, the paths are credited against deliberate attack only when they are robust, diverse and independent: two or more paths, differing in kind, and sharing no location, power supply, control system, vendor or update path, so that no single act defeats them all. Gates driven from one control house, or two warning paths through one communications building, are not credited against deliberate attack (*Defense in Depth* §3.4). An ungated overflow crest or a fuse plug, which passes water by its own height and needs no power or command, is the clearest independent path the class has. Where the dam cannot show independent release capacity and warning against every combination of losses that would defeat them, the gap must be stated openly as a declared residual, in the form *Defense in Depth* §3.5 sets out (Section 6.8). Discharge through a hydroelectric plant's turbines is not such a path, because it depends on the generating plant (Section 5.1).

Design-basis duration event-specific; for release capacity, the duration of the inflow it must pass.

5.7 Upstream Release in a Cascade

Postulated event the failure of, or an uncontrolled release from, a dam upstream, delivering a flood the dam was not designed to receive.

Consequence threshold overtopping or failure of the dam under the arriving flood, adding its reservoir to the release (Chain I), or the loss of its pool to the scour of the arriving flood (Chain P).

Required function every dam in a cascade must state the cascade loading of Section 3.3 and either pass it, survive its overtopping, or carry, as a declared residual, the fact that it will fail when the dam above fails. The Emergency Action Plans of the dams in a cascade are written as one plan for the warning: the warning time downstream of the lowest dam is counted from the first sign of failure at the highest dam that can fail, and the downstream population is warned on that sign, not on the failure of the dam nearest to it. Where the dams of a cascade are regulated by different authorities, the design basis states who holds the cascade's analysis and its warning.

Design-basis duration from the first sign of failure upstream until the arriving flood has passed the lowest dam.

This category is characterized empirically by the failures of the Edenville and Sanford dams on the Tittabawassee River on 19 May 2020, conferred by the framework as DBA-DAMS-RE-3 (framework §9.1). Four dams of one owner stood in a cascade, from Secord and Smallwood above to Edenville and Sanford below. In May 2020 Secord, Smallwood and Sanford were licensed by the Federal Energy Regulatory Commission, and Edenville, its license revoked in 2018, was regulated by the state, on the Independent Forensic Team's interim account. Edenville failed and Sanford, overtopped by the flood, failed after it. About 11,000 people were evacuated, on the Federal Emergency Management Agency's incident response review, and the state reported no fatalities. The county ordered evacuation before the Emergency Action Plan was formally activated, on the accounts of that review and of the press. The design-basis reading is that the cascade behaved as a single system while its dams were regulated separately, and that the warning time downstream was set by the first failure, not by the nearest one. That the transfer of Edenville's regulation from the federal to the state program left the state with little information about the dam, a matter the Forensic Team's final report addresses, is for the reference event to examine.

5.8 Seismic and Other Natural-Hazard Exceedance

Postulated event earthquake, landslide into the reservoir, extreme cold or ice, or wildfire exceeding the design parameters of the structure, its works or their power and access.

Consequence threshold a breach or a condition that will become one (Section 5.2), or the loss of release capacity or warning (Sections 5.4 and 5.6).

Required function the structure, its works and their backups must remain within their design basis under the design-basis natural hazard, with particular attention to a hazard that takes a gate's power, its access and its backup together. The dam also states what it does when the hazard exceeds that basis: the margin between the design-basis hazard and the point at which the structure or its release capacity is lost, and the declared residual beyond it, in the output form of *Defense in Depth* §3.5. The design basis also postulates a significant inflow arriving after an earthquake and before the structure has been shown stable, within the interval the dam's post-earthquake inspection plan allows for that showing. During that interval the structure is not credited with the strength it had before the earthquake, and a release path or gate whose condition has not been confirmed is not credited in the release capacity. The design basis states in advance the pool level at which the dam is held through the interval, the release capacity it credits, and the inflow at which it lowers the pool further or warns the valley. The length of the interval is the site's, set by its inspection plan, and is not fixed here. The compound event is taken in from *Large Hydroelectric Storage*, which postulated it before this series existed (Section 10.3).

Design-basis duration event-specific; for an earthquake, until the structure is shown stable after the event; for an inflow arriving before then, until the inflow has passed and the structure is shown stable.

Table 5-1. Facility-class design-basis event categories.

Event category	Postulated event (class-specific)	Consequence chain	Framework source
Hydrologic loading beyond release capacity	Inflow beyond the works at the gates available; waves beyond the freeboard	Chain I	§4 (hydrologic case); developed here
Structural or foundation failure	Sliding, internal erosion, static liquefaction, cracking	Chain I	§4 (structural case); developed here
Failure of a spillway or outlet works	Chute damage, erosion below an emergency spillway, conduit failure	Chain I	§4.1 candidate (uncontrolled gate or spillway release)
Loss of gate power, function or control	Loss of gate or instrument power, gate failure, erroneous command	Chain I + Chain P	§4.1 candidate; §5

Event category	Postulated event (class-specific)	Consequence chain	Framework source
Cyber attack on gate control and monitoring	Gate commanded, indication falsified, failure hidden	Chain I	§5; developed here
Physical attack on critical nodes	Hoists, control house, gate power, monitoring, alarm path	Chain I	developed here
Upstream release in a cascade	Failure or release of a dam upstream	Chain I + Chain P	developed here
Seismic and natural-hazard exceedance	Earthquake, landslide, ice, wildfire beyond design; inflow before post-earthquake clearance	Chain I	developed here

6. Military-Action Extension: DBA-MA for the Dam Structure

6.1 MA Extension Applicability Determination

Every dam in the class performs the facility-level applicability determination the series makes mandatory (framework §7). The primary factor is geolocation relative to military-action exposure, and the secondary factor is strategic value. The determination is time-varying and re-triggered on the geopolitical indicators the DBA-MA parent framework defines. The framework states a strong sector-level presumption toward applicability for every high-hazard dam (framework §7). For this class, strategic value is driven by three things: the population in the deliberate-case footprint of Section 4.5, the dependents of the pool under Chain P, including any nuclear or other facility whose safety relies on it, and membership in a cascade whose failure the dam would begin or join. The determination is recorded, and any scope reduction carries a recorded basis (Section 6.9).

6.2 The Four Characteristics Applied to the Class

Applied to the dam, the four military-action characteristics take specific form. Intentionality: the adversary selects the dam because of the flood it holds and the dependents that rely on its pool, and the dam's purpose of keeping energy in reserve becomes the reason it is chosen. Persistence: the adversary holds a structure, or access to its controls, across the interval needed to choose the moment; Kakhovka was held by Russian forces from the opening days of the full-scale invasion in February 2022 until its destruction, as widely reported, and the Bowman Avenue intrusion was repeated over three weeks, on the Justice Department's account. Persistence also acts through the denial of the dam's maintenance and inspection over months. A structure its owner cannot inspect is one whose deterioration goes unseen, and the stability Section 5.2 requires to be shown cannot then be shown. Under a positive applicability determination the design basis states the longest interval without inspection after which the structure is no longer credited with the

stability last shown, and the pool level at which the dam is held from then on. Precision: the adversary targets the section of the structure, the gate or the power supply whose loss produces the governing consequence, and chooses the pool level, the state of the river and the hour that carry the flood farthest and give the least warning. Denial: the adversary disables or blinds the monitoring and the warning before acting, denies the site to the crews who would operate the gates by hand, and acts during a conflict in which the response downstream is itself under attack. At Kakhovka evacuations from the flooded areas came under shelling, each side's officials attributing it to the other, and the United Nations said it could not reach the areas under Russian control. The class validation standard is the state actor who holds or can reach the dam, knows what its flood will do downstream and what its pool sustains upstream, and has chosen the day.

6.3 The Deliberate Breach as the Governing MA Scenario

The governing military-action scenario is the deliberate breach of a high-hazard dam at a high pool, with the warning denied. It opens both chains at once. The release is counted as Section 4.2 requires: at the highest permitted pool, with the breach forming as fast as the means can make it, at every place the act can open, and through every dam below that the flood can take. No credit is taken for warning that the same act or campaign can deny. The extension therefore requires three things. It requires a warning that works without the dam: the first sign of a breach detected by means off the structure and independent of its owner's systems, and the warning carried to the downstream population by paths the act cannot cut. Under a positive applicability determination, the loss together of the dam's detection and its warning paths is a sign of a release for the purpose of the warning time, and the warning goes out on it (Section 4.5). It requires an evacuation plan for the deliberate-case footprint, not only the accidental one, that does not rely on an off-site response the same campaign may be denying. And it requires a pool-management rule under threat: the pool level at which the dam is held when the applicability determination is positive, stated in advance with the authority that orders it, the cost to the dependents of Chain P that it carries, and the margin it buys downstream. The authority weighs the margin downstream against the margins the dependents' own design bases provide independently of the pool, not against their needs as stated, so that a dependent with no reserve of its own does not hold the drawdown back by default; a dependent without such a reserve is a finding for its own design basis, to which the pool-dependency margin is given (Section 8.1). Where the dependent is a nuclear plant, its margin is the one its own design basis states for the loss of its cooling water (Section 10.9). DBA-DAMS-UC-1 (*A Deliberate Breach at High Pool with the Warning Denied*) drives this scenario through the class (Section 9.2).

This scenario is characterized empirically by the destruction of the Kakhovka dam on 6 June 2023, which the framework names as the sector's reference event (framework §1.5, §7) and which the framework confers as DBA-DAMS-RE-1 (framework §9.1). The Norwegian seismic monitoring institute NORSAR recorded signals at 02:35 and 02:54 local time that it found consistent with an explosion and not with what would be expected of a dam collapse; it did not determine the device. The Government of Ukraine stated that Russian forces destroyed the dam.

The Government of the Russian Federation stated that it was deliberate sabotage by the Ukrainian side. The Secretary-General of the United Nations called it a monumental humanitarian, economic and ecological catastrophe and said the United Nations had no independent information on how it happened. This instrument makes no finding of attribution. Deaths were reported by each side's authorities and by later independent reporting, and the counts are contested; the reference event, not this class, will state them, each attributed to whoever published it. The design-basis reading is that a dam can be destroyed as an act of war, that its flood and the loss of its pool arrived together, and that the upstream dependent nearest to a nuclear consequence survived on the margin of its own cooling pond, not on any provision of the dam.

6.4 The Coordinated Cyber Scenario

The governing cyber scenario is what the Bowman Avenue intrusion would have done had the gate been connected, at a dam whose gates can release a flood. The gates are commanded open at a high pool, or held closed during a flood, while the operators see normal readings and the monitoring of the structure shows nothing. Against this scenario the class relies on the governed-change requirement of Section 5.4, on independent indication of gate position and pool level (Sections 5.4 and 5.5), and on release capacity that no network can reach: an ungated crest or a fuse plug that passes water by its height (Section 7.2). Operator detection cannot be assumed and displayed values cannot be trusted. The scenario has a second form that needs no gate at all: a falsified reading that leads the operators to release, or not to release, on their own authority. That form is governed by the integrity of the basis on which a release is decided, and the obligation is FD-BL-D2's (Section 7.3). DBA-DAMS-UC-2 (*Gates Opened at High Pool Behind Normal Readings*) drives both forms through the class (Section 9.2). Against the first, beside the governed act, the class relies on the rate limit of Section 5.4, which bounds how fast a commanded gate can raise the river. Against the second, it relies on the continuous comparison of independent indication with the operators' indication that Sections 5.4 and 5.5 require, because the divergence of the two pictures is the only sign a clean falsification gives.

6.5 The Inundation Map as a Targeting Aid

The inundation map that tells a downstream population where to go also tells an attacker what a breach would do, and the hazard rating that marks a dam for inspection marks it for a target. Article 56 of Additional Protocol I to the Geneva Conventions provides an optional marking of three bright orange circles for works containing dangerous forces, which identifies the work to every party. The design-basis response follows the chemical framework's for hazard labels (*Chemical Sector Framework* §7.1): the information stays, because the population and the responders need it, and the dam's design basis assumes the adversary has read it. Whether particular inundation detail should be withheld from public release is a question for the regulators who hold it and is not decided here. Withholding detail from public release does not

withhold it from the facilities in the footprint, which receive the figures for their own sites under Section 8.1.

6.6 Cross-Sector Interaction

Under military action the dam is tied to other sectors in both directions. Coming in, an attacker may time an act to a loss of grid power, which takes the gates' main supply, or to a loss of communications, which takes the alarm, or to a flood already in progress (Sections 5.4, 5.6 and 4.3). Going out, the flood reaches the facilities in its footprint, chemical plants, water and wastewater plants, substations, bridges and the stations of the responders who would act, and the loss of the pool reaches every dependent of Chain P. The Kakhovka coupling is the sharpest: a nuclear plant on the shore of the reservoir whose cooling relied on it. The adversary need not choose between the chains; the same act opens both.

6.7 The Adequacy Result: Autonomous Safe Impoundment Condition

The test of whether a dam's protection is adequate is whether it can reach what this document calls the Autonomous Safe Impoundment Condition. When networked control, grid power and outside help are lost, the dam holds its pool within the band that keeps the structure safe and passes the inflow without overtopping, on its own. The structure carries its loads; the release capacity the design basis credits is available without outside power or network command, through gates that can be worked locally or through an ungated crest that needs no one; every gate is in a position decided in advance for that condition, and it cannot be moved out of it except by a governed act; and the first sign of failure reaches the downstream population by a path that does not depend on the dam's own systems.

The protective functions on which this condition rests are to be locally sovereign as FD-LS defines the property (*Local Sovereignty* §3.2). The decision is a measurement against a setpoint, or no decision at all, as with water passing over a crest at its own height; the function is colocated with the structure it protects; no system an attacker could reach can stop, delay or trigger its action; and no such system can alter its settings or its state except through a governed act. Local Sovereignty governs and is not restated here. Local sovereignty describes what protects the dam when its control systems and the authorization architecture around them have failed. It does not decide whether any action needs human authorization (*Local Sovereignty* §4).

A small crew on site must also be able to hold this condition for a set period with no outside power, communications or support. That requirement is written here in this document's own words. Other instruments take a similar requirement from the DBA-MA parent framework's "local sovereignty principle", but whether that principle is the same thing FD-LS defines is still an open question, so this document does not borrow it (Section 10.5). The crew's endurance is bounded by its own safety: a crew cannot be counted on to work the gates of a dam under fire, or one that is failing beneath it. Where the condition depends on people reaching the gates, the condition states how long it holds without them.

The end state may also be reached by lowering the pool. A drawdown taken in advance reduces the stored energy, and with it the flood a breach would make, at a cost to every dependent of Chain P. It is a protective action like an evacuation, and like an evacuation it needs an authority named in advance to order it, a trigger stated in advance, and a statement of what it costs and to whom. The class requires the Emergency Action Plan, or the plan that stands beside it under threat, to name that authority (Section 6.3). Where the authority named is not at the dam, the plan states whether the crew may order the drawdown when the authority cannot be reached. Where it does not so delegate, a drawdown under denial rests on the pool rule of Section 6.3, stated and ordered in advance of the denial (Section 8.5).

6.8 Adequacy and the Declared Residual

In the non-adversarial domain, the Autonomous Safe Impoundment Condition is achievable with the Section 5 requirements: release capacity at the gates as they can be worked, stability shown along the whole structure, release paths proven, gates and warning independent of the grid and the network. Against a deliberate attacker, adequacy has limits. Against a cyber attack, it rests on gates that can be moved only through a governed act, on independent indication, and on release capacity that no network can reach. Against a deliberate breach, it rests on warning, evacuation and the level at which the pool is held, because a structure built to hold back a river cannot also be built to withstand every means of destroying it.

Three residuals are declared, in the output form of *Defense in Depth* §3.5. First, a sufficiently large deliberate act can breach any dam in the class, however good its cyber defenses; beyond the design act the residual is a flood the dam cannot prevent, and the response is warning, evacuation and a pool held low in advance, not autonomy. Second, sustained denial of site access during a high inflow can outlast any release capacity that depends on people reaching the gates; beyond that point the residual is an overtopping the dam cannot prevent, and the design response is passive release capacity sized for the inflow the denial period can bring. The same residual, and the passive capacity sized against it, cover an overtopping under a falsified basis, with the crew present and misled about the pool or the gates, as well as one under denial; the passive capacity needs no reading and no decision, and it is sized for the inflow either can bring. Third, in a cascade, a dam that cannot pass the failure of the dam above carries that failure as its own residual (Section 5.7). These residuals bound the class adequacy claim.

6.9 Consequence-Basis Conditionality Check

Any relaxation of a conservative class parameter, such as a pool level below the highest permitted, a slower breach, a longer warning time credited or a cascade loading set aside, must be justified against the class baseline, and the larger the relaxation the stronger the analytical basis required. A relaxation without a recorded basis is not consistent with the class design basis. The framework does not yet state this principle for the sector, and Section 9.1 records it for the framework to adopt, as the transportation framework has.

7. The Dual Design Basis, Stated in Full

7.1 The Two Design Bases

The dam carries two design bases at once: a physical design basis, defined in this instrument, and an AI-governance design basis, governed by the Foundational Definitions and the Command Broker family and named and referenced here (framework §5). They interact at the point developed in Section 7.4.

7.2 The Physical Design Basis: The Determinism Posture

The physical design basis is the set of protective functions the dam relies on to hold its safe state independently of the networked control system. These are the structure itself, the ungated overflow crest and the fuse plug, the gates and their local and manual operation, the mechanical limits on gate travel and on the rate at which a gate opens (Section 5.4), and the integrity of the indication of pool level, gate position and the condition of the structure. Water passing over an ungated crest is the clearest locally sovereign function in the class: it is physics, at the structure, and reachable by no network. A fuse plug, built to wash out at a set level and open a further path, is the same in kind. A gate meets *Local Sovereignty* §3.2 only if it can be worked at the structure without any system an attacker could reach, and its position cannot be changed from such a system except through a governed act. A gate that can be moved from a network an attacker can reach fails that test. The Bowman Avenue gate failed it, and was saved by a maintenance outage (Section 5.5).

Whether a function is deterministic decides whether the authorization rules apply to it at all. It does not decide where its actions fall on the Bright Line. Gate controllers and level-control loops that behave predictably within a stated range of inputs, and that have been formally tested as installed, are exempt from the Command Broker requirement for the actions they take while their inputs stay in that range (*Layer Decomposition and Determinism* §4.3). They are not below the line: they take some of the highest-consequence actions at the dam, and they are outside the obligation because their behavior is bounded by proof (*The Bright Line* §4.5). The exemption rests on the configuration as installed and tested, and it lapses for a configuration changed outside a governed act, as the production class states for its safety controllers (*Chemical Production and Synthesis* §7.2). A component that has not been shown to meet the test is covered by the rules, and each of its actions is judged on its consequence. This is stated for the class's physical design basis and does not vary the scope gate that *Layer Decomposition and Determinism* defines.

7.3 The AI-Governance Design Basis: Named and Referenced

Machine-learning and generative-AI systems at the dam — reservoir-operations optimizers, inflow forecasting, automated gate scheduling, machine-learning analysis of the structure's instruments, and operator decision support — are governed by the Bright Line and Command Broker architecture (framework §5) and by the Command Broker family, whose parent is CB-

Framework_Cross_Sector (*Command Broker Framework*). CB-REG (*Command Broker Family Register*) is authoritative for the family's version state. The framework puts responsibility for applying that architecture with the coordination class, OpsCo (framework §3.5). This document names the actions at the dam that the architecture applies to. Placement is by the consequence of each action, not by the component (*The Bright Line* §4.2, §4.3).

Three cases illustrate the application, and none of them is a placement by component. An optimizer within scope that schedules gate movements takes an action with every command it issues, and each command is placed on its consequence. A command that could release a flood reaching the endpoint downstream, or hold a gate closed while the pool rises toward the crest, or draw the pool below the level a dependent needs, is above the line and requires Command Broker authorization; a command whose consequence cannot reach those conditions is placed below it. The framework's statement that a gate opening at a high reservoir level must sit above the line is read in this sense: it names the consequence that places such a command, and a gate command at a low pool whose release cannot reach the endpoint is placed on its own consequence (framework §5). A generative assistant is within scope: a release plan it proposes, together with the operator's acceptance of it, is one action placed on the consequence of the release. A machine-learning model that reads the structure's instruments and decides whether a reading is an alarm is within scope where it does not meet the exemption test, and any output on which a decision to release, to hold, or to warn may be grounded carries the indication-integrity obligation of *Indication Integrity*. A model that suppresses an alarm removes the warning on which the valley depends, and its output is placed on that consequence.

7.4 The Interaction Point: The Safe Impoundment Under Denial

The two design bases meet when the dam must hold its pool safely while under attack. The control system must be architecturally incapable of executing an above-line action originating from a component within the scope of *Layer Decomposition and Determinism* §4.3 without Command Broker authorization. That includes any command that moves a gate toward a release reaching the endpoint downstream, holds a gate closed against a rising pool, or draws the pool below the level its dependents need. And the safe condition must hold even when no outside system or authority can be reached at all. The physical design basis guarantees the safe state; the AI-governance design basis guarantees that no proposing system can move the dam out of it without a qualified human authorization the architecture can enforce.

7.5 Scope Boundary

The physical design basis is defined in full in this instrument. The AI-governance design basis is named and referenced, not redefined; its governing text is *The Bright Line* with its determinations, and the Command Broker family under its register. Where the two are cited together, the physical requirement governs the dam's protective architecture and the AI-governance requirement governs the authorization of any proposing system that interacts with it.

8. Cross-Sector Dependencies and Regulatory Context

8.1 What Other Sectors Depend On: the Pool and the Valley

Two sets of facilities depend on the dam. The first are the dependents of the pool: the dam's own generation (*Large Hydroelectric Storage*), the water supply drawn from the reservoir (*Storage Reservoir*), navigation through the reservoir and its locks (*Ways and Linehaul*), irrigation, and any plant that draws cooling water from the reservoir. Each depends on the pool staying above the level it needs, and Chain P is measured against its reserve. Where such a dependent is rated Tier 2 or Tier 3 in its own series, the pool-dependency margin of Section 3.3 is stated for it at the dam and given to it, so that its own design basis can count it. The second are the facilities in the footprint of Chain I, whose own design bases should count the dam above them as a source of flood. The class requires that every facility within the deliberate-case footprint of Section 4.5 receive, under whatever handling regime applies to the inundation detail, the depth, speed and arrival time of the flood at its own site, for the deliberate case as well as the accidental ones, so that it can design against the map the adversary is assumed to have read (Section 6.5). The dam's design basis is not met until each such facility has received them. The regulators' decision on the public release of inundation detail governs what is published, not what the facilities in the footprint are given.

8.2 What the Dam Depends On

The dam depends on electric power, communications, forecasting and access (framework §6.1). Electric power moves the gates, and its loss beyond the endurance of backup is a design-basis condition of Section 5.4. Communications carry the alarm, and the framework names the warning path as the most consequence-critical dependency, because the Emergency Action Plan's warning fails if its notification path fails (framework §6.1); the communications design basis applies to it, and Section 8.5 states how. Forecasts of precipitation and inflow set the decision margin of Section 3.3, and their loss shortens it. Access lets crews reach the gates and the structure. A dam can also depend on another dam: the one above it in a cascade, whose release is its inflow (Section 5.7).

8.3 The Boundaries with the Partner Series

This instrument states the boundaries the framework names (framework §6.3), on its rule that the served function is the partner's and the structure and its release are this series'. With *Large Hydroelectric Storage*, the powerhouse, the turbines and the generating consequence are the electric series', and the structure, the spillway, the gates and their controls, and the release of the impoundment through any path, including a penstock, are this class's. That instrument was written before this series existed and carries a dam-safety consequence chain of its own. On the author's determination of 12 October 2026, recorded in the return note Project 8 sent this project on 12 October 2026, that chain has moved here. *Large Hydroelectric Storage* keeps its generating chain and the interaction between the generating chain and the dam-safety chain,

which is its own: its dual-chain events, its combined consequence assessment and its black-start question (*Large Hydroelectric Storage* §4.2, §7.4, §8.4). Its analysis of the dam-safety chain has been replaced by citation to this class (Section 10.3). A penstock rupture that loses generation is the electric series' consequence; the same rupture as an uncontrolled release of the reservoir is this class's. With *Storage Reservoir*, the structural and impoundment safety case of a regulated dam is this series', as that instrument already states, and the water-supply function is the water series'. With *Ways and Linehaul*, the lock chamber, its gates and valves and their machinery are the transportation series', and the dam, the impoundment and their release are this series'.

8.4 Regulatory and Institutional Context

The class sits under an established dam-safety apparatus (framework §2.1). For non-federal hydroelectric dams the Federal Energy Regulatory Commission's regulations on the safety of water power projects govern, at 18 CFR Part 12. Subpart C requires Emergency Action Plans (§12.20 to §12.25). Subpart D requires inspection by an independent consultant, which since the Commission's Order No. 880, effective in April 2022, alternates every five years between a periodic inspection and a comprehensive assessment that reviews the design basis, the analyses of record and spillway adequacy, with a potential failure modes analysis and a risk analysis; the same order codified the requirement for an owner's dam safety program. Federal dams are governed by the agencies that own them, the Army Corps of Engineers and the Bureau of Reclamation among them, under the *Federal Guidelines for Dam Safety*, and most dams are regulated by the states. The class requires that a dam's Emergency Action Plan inundation analysis and its potential failure modes analysis be consistent with Section 4. The guidance the class follows for inundation mapping is FEMA P-64 and FEMA P-946, for hazard potential FEMA P-333, and for risk FEMA P-1025.

Security is less settled than safety. The Commission's *Security Program for Hydropower Projects*, in its Revision 3A, places licensed dams in three security groups according to hazard classification, size, consequence and generating capacity, and has a section on cyber assets; it is guidance enforced through the Commission's inspections, not a regulation. The Cybersecurity and Infrastructure Security Agency's Dams Sector guidance is voluntary. A dam outside the Commission's jurisdiction has no federal security requirement that this instrument has found, and the class records that as a gap to be confirmed at each revision (Section 10.8). Article 56 of Additional Protocol I to the Geneva Conventions provides that dams shall not be made the object of attack, even where they are military objectives, if the attack may cause the release of dangerous forces and consequent severe losses among the civilian population, subject to narrow exceptions. The United States signed the Protocol in 1977 and has not ratified it, and in 1987 the State Department's Deputy Legal Adviser stated that the United States did not support the provisions of Article 56 or consider them customary law. The class does not rely on the law of armed conflict to protect a dam; it records the law as part of the setting of the military-action extension.

8.5 The Communications Design Basis Applied to the Dam

The communications design basis is cited, not inherited. DBA-MA-CDB (*Military Action Communications Design Basis*) sets out seven requirements, CDB-1 to CDB-7, for every facility whose communications may be degraded or denied, and its §3 requires each facility-class instrument to state how they apply. This section states how the seven apply to the dam and its works; their text is in that document and is not restated. The dam-safety obligations that *Large Hydroelectric Storage* states for its class come here with the dam-safety chain (Section 10.3): the activation of the Emergency Action Plan, the operation of the gates, the ordering of a drawdown, the warning of the valley by alternate means, and the channel to the dam-safety authority. That instrument keeps the communications plant of a hydroelectric dam, which serves both its grid duties and the duties this class governs, and states its own obligations for the grid.

A dam is often remote, and the people who must act when it is threatened are few and spread apart: the crew at the dam, the owner's engineers, the dam-safety authority, the emergency managers who warn and evacuate the valley, and, in a cascade, the operators of the dams above and below. An adversary who means to breach a dam or misuse its gates silences it first (Section 4.5), so the communications the dam needs most are the ones the act is designed to remove. The requirements below follow from that.

CDB-1, degradation as baseline. The design basis assumes from the outset that the dam's supervisory and voice links, to its owner's control center, to the dam-safety authority and to the emergency managers downstream, are degraded or lost in any event large enough to matter. The Emergency Action Plan, the pool rule of Section 6.3 and the warning of the valley are written and exercised against that condition.

CDB-2, local protective and safety autonomy. This requirement is still worded for the electric sector, where it names the supervisory and energy-management systems. Read for the dam, it requires that loss of the gates' supervisory control or of the structure's monitoring network not cause loss of the local operation of the gates, the mechanical limits on their travel and on their rate of opening, or the passive release capacity of the crest and the fuse plug. Section 7.2 states the property, and the design basis must demonstrate it rather than assume it. The crew at the dam is authorized in advance, in the Emergency Action Plan, to activate the plan and to operate the gates under its procedures without outside contact. A drawdown is ordered by the authority the Emergency Action Plan names (Section 6.7), and the crew may order one only where the plan delegates it; where it does not, a drawdown under denial rests on the pool rule of Section 6.3, stated and ordered in advance. Authorization in advance settles what the crew may decide on its own; it does not change how a gate is moved, which remains a governed act (Section 5.4).

CDB-3, field-executable emergency procedures. From procedures and controls held at the dam that do not depend on the network, the crew must be able to operate by local or manual means every gate the release capacity credits, leave each in the position decided in advance for the Autonomous Safe Impoundment Condition (Section 6.7), activate the Emergency Action Plan,

and start the warning of the valley by alternate means, such as sirens, runners and the local emergency managers, when every electronic path is lost. The procedures state the threshold at which the warning goes out by those means, and under a positive applicability determination that threshold includes the loss together of the dam's detection and warning paths (Section 4.5).

CDB-4, verified independent channels. The dam keeps at least one channel to its dam-safety authority, whether the Federal Energy Regulatory Commission, the Corps of Engineers, the Bureau of Reclamation or the state, and at least one to the emergency managers who warn the valley, each sharing nothing with the dam's primary supervisory and voice networks: licensed radio, satellite, or physically diverse fiber. The warning of a release is carried by paths the act cannot cut (Section 6.3), and the detection off the structure that starts it is OpsCo's (Section 10.6). In a cascade the dams share a channel of the same kind, so that the first sign of failure at the highest dam that can fail reaches every dam and every population below it (Section 5.7). At a hydroelectric dam the channel to the control center that dispatches the plant is the electric series' and is stated in *Large Hydroelectric Storage*; the channels serve different duties, and each meets this requirement on its own.

CDB-5, coordination without shared connectivity. This requirement is also still worded for the electric sector, where it names the gas–electric boundary. Read for the dam, it requires that the owner, the dam-safety authority, the emergency managers, the dependents of the pool and, in a cascade, the owners of the other dams be able to carry out the single warning plan of the cascade (Section 5.7) and the drawdown decision of Section 6.3 when they share no network, under a plan that names who decides what and how they tell each other.

CDB-6, Command Broker continuity. The loss or degradation of the dam's communications does not suspend the Command Broker obligation. CDB-6 states the obligation for actions recommended by generative AI; this instrument applies it to every component within the scope of Layer Decomposition and Determinism §4.3, which is the scope the Foundational Definitions set, and Sections 7.3 and 7.4 apply it to the gate commands, release plans and alarm decisions of the dam.

CDB-7, verification cadence. The independent channels of CDB-4 are tested at intervals the design basis states, with the primary supervisory and voice links assumed lost. The local and manual operation of the gates, the crew's activation of the Emergency Action Plan and the warning of the valley by alternate means are exercised at the same intervals under the same condition, and the time from the first sign of a release to the warning of the nearest population is measured in each exercise, because that time is the warning time of Section 3.3.

9. Inheritance Map and Anticipated Derivatives

9.1 The Inheritance Map

The series requires every facility-class instrument to state, at first publication, which elements it inherits unchanged, which it inherits with adaptation, and which it defines locally. Facility-class

instruments inherit the framework's methodology and architecture and add their structure-specific consequence chain and boundaries (framework §8.1); Table 9-1 states that inheritance element by element. Where the framework is a skeleton, the class content is developed ahead of it and is marked for the framework to take up.

Table 9-1. Inheritance map for DBA-DAMS-DS-FC1.

Element	Source in the framework	Inheritance mode
Function-primary placement, DS function class	§1.3, §3.1, §3.2	Inherited unchanged
Hazard-potential sub-index (FEMA P-333)	§2.2, §3.1	Inherited with adaptation; deliberate-case rating added (Section 4.5)
Design-basis principles (consequence severity, not likelihood)	§4.1	Inherited unchanged
Inundation consequence measure	§3.6, §9 (requirement)	Developed here as the provisional home (Section 4)
Event categories	§4.1 (candidates)	Developed here ahead of the framework; for the framework to take up
Bright Line and Command Broker architecture	§5; FD-BL; Command Broker family	Inherited with adaptation: the gate-opening statement read by consequence (Section 7.3), for the framework to confirm
Inward and outward dependencies; named boundaries	§6.1, §6.2, §6.3	Inherited with adaptation; Chain P and the boundaries stated (Sections 3.2, 8.3)
MA applicability and its presumption	§7	Inherited with adaptation to facility level
Hazard information as a targeting aid	not stated in the framework	Adapted from the chemical framework; for the framework to take up
Local sovereignty of protective functions	FD-LS §3.2 (not the framework)	Cited directly
Communications design basis	DBA-MA-CDB §2, §3 (not the framework)	Cited directly; applied to the class at Section 8.5
Independent protection and declared residual	DBA-MA-DiD §3.4, §3.5 (not the framework)	Cited directly; for the framework to take up

Element	Source in the framework	Inheritance mode
Consequence-basis conditionality check	not stated in the framework	Locally stated; for the framework to adopt
Consequence chains, design parameters and the Autonomous Safe Impoundment Condition	—	Locally defined

9.2 Anticipated Derivatives

Site-, state- and contract-specific dam design bases inherit this class document by reference and add only what is locally specific, each carrying a recorded applicability determination for any scope it reduces. A cascade design basis, written for the dams of one river together, is anticipated beneath this class where the dams of a cascade have different owners or regulators, and its designator is the framework's to confer. The four reference events this class proposed at v0.1 are conferred by the framework at §9.1 on the author's decision of 12 October 2026: DBA-DAMS-RE-1, the destruction of the Kakhovka dam, June 2023 (Section 6.3); DBA-DAMS-RE-2, the Oroville spillway incident, February 2017 (Section 5.3); DBA-DAMS-RE-3, the failures of the Edenville and Sanford dams, May 2020 (Section 5.7); and DBA-DAMS-RE-4, the intrusion into the Bowman Avenue Dam's control system, 2013 (Section 5.5). Each will read its event back through this class's design basis, and their instruments are Project 11's to write. Three use cases have issued under this class: DBA-DAMS-UC-1 (*A Deliberate Breach at High Pool with the Warning Denied*) for the governing scenario of Section 6.3, DBA-DAMS-UC-2 (*Gates Opened at High Pool Behind Normal Readings*) for the cyber scenario of Section 6.4, and DBA-DAMS-UC-3 (*The Facilities in a Dam's Flood Path*), which follows one release through the classes of four series in the footprint. They return eight gaps to this class, and this revision takes in all eight, at Sections 4.5, 5.4, 5.5, 6.3, 6.4, 6.5, 6.8, 8.1 and 10.6. The findings they note to the classes of other series are recorded at Section 10.9.

10. Gaps and Open Items

10.1 Quantitative Design Parameters

The governing release, inundation footprint, warning time, release capacity, decision margin, cascade loading and pool-dependency margin of Section 3.3, and the endpoint of Section 4.4, are stated as parameters, not numbers. Their values await validation with dam owners against the inundation analyses, potential failure modes analyses and spillway adequacy studies they already hold (Section 8.4). Where those analyses take the pool at its normal level only, or a breach that forms at an accident's pace, validation against them does not validate the deliberate case, which Section 4 sets.

10.2 Design-Basis Durations

The framework's design basis is a skeleton (framework §4) and sets no durations. The durations in Section 5 are stated as class parameters, and the class durations for holding the safe impoundment without outside power, communications and support await the framework's values. With the move of the dam-safety chain (Section 10.3), the impoundment half of the military-action acceptance criterion of *Large Hydroelectric Storage* comes to this class, and the safe shutdown of the generating units stays with that instrument. That instrument sets 72 hours without on-site personnel; Project 8 offers the figure to the framework as a value already in the corpus, not as one it sets for this series. It is recorded here for the framework to weigh, and this class takes its duration from the framework when the framework states one.

10.3 The Boundary with the Hydroelectric Class

The boundary with *Large Hydroelectric Storage* is closed. That instrument carried a dam-safety consequence chain written before this series existed (Section 8.3), and the framework intends the dam-safety consequence to be consolidated here (framework §1.5, §6.3). This project raised the boundary with Project 8 on 12 October 2026, and on the author's determination of the same day, recorded in the return note Project 8 sent this project on 12 October 2026, the chain moved. *Large Hydroelectric Storage* has since issued the revision that carries the move, recorded in the notice Project 8 sent this project on 12 October 2026. It keeps its generating chain and the interaction between the two chains, and it cites this class for the dam-safety chain. This project read every one of its citations into this class against this class's text. Each lands on the section that states what it cites. Three found gaps in this class, which this revision fills: waves beyond the freeboard (Section 5.1), the loss of power to the instruments on the structure (Section 5.4), and a long denial of inspection (Section 6.2). The drifts Project 8 found between the two instruments before the move are not carried here: the governing case for the dam-safety chain is the deliberate breach of Section 6.3, with sustained denial during a high inflow the second residual of Section 6.8; a gate opened when it should not be is a Chain I release, and a release that draws the pool below what its dependents need is Chain P, whatever the pool (Section 5.4); and gate commands are placed on the consequence of each command, with exempt deterministic controllers not below the line (Sections 7.2 and 7.3). The three items the move offered were taken in at Version 0.3: turbine discharge as a release path (Section 5.1), the earthquake followed by an inflow before the structure is shown stable (Section 5.8), and the dam-safety communications obligations (Section 8.5).

10.4 The Inundation Model's Provisional Home

Section 4 holds the shared inundation consequence model provisionally, on the author's decision of 12 October 2026, as the chemical-release model is held in *Chemical Production and Synthesis* §4. When the framework confers a designation on a standalone model instrument, Section 4 moves there, this instrument cites it, and every instrument citing Section 4 in the meantime is owed notice that a sweep is due. The prospective citers are the series' other classes, *Reservoirs*

and Appurtenant Works, DBA-DAMS-LV-FC3 and DBA-DAMS-OpsCo, when they are written, and, outside the series, *Storage Reservoir* at its structural chain. *Large Hydroelectric Storage* already cites it, having closed its downstream flood consequence mapping item by citing Section 4, and is owed the notice as well. The item is registered under Project 9 OI-32, beside OI-18 for the chemical-release model.

10.5 Local Sovereignty in the DBA-MA Sense

The protective-function property at Section 6.7 cites *Local Sovereignty* §3.2 directly. The endurance requirement beside it, holding the safe impoundment without outside power, communications or support, is the property the DBA-MA parent framework's local-sovereignty principle describes. Whether that principle is FD-LS's construct at all is held pending Project 5's answer, and the requirement is stated locally until then. The item is registered under Project 9 OI-16.

10.6 The Boundary with the Coordination Class

The framework places the SCADA, reservoir-level and seismic monitoring systems in OpsCo, as the sector's detection-and-warning layer (framework §3.5). This instrument proposes, for the framework to decide, that the instruments mounted on the structure and the gates' own controls are this class's hardware, and that the function of turning a reading into a decision and a warning is OpsCo's. The rule is offered for the framework to adopt or replace, and the boundary is listed as an open item until it does. On the finding of DBA-DAMS-UC-1, the proposal extends to detection off the structure: the detection of a release by means off the structure and independent of the owner's systems, which Section 6.3 requires, is OpsCo monitoring and not this class's hardware, so that the single warning plan of a cascade (Section 5.7) has a detection to start from. This class requires that detection; OpsCo holds it.

10.7 Reference Events and the Use Case

The four reference events this class proposed at v0.1 are conferred by the framework at §9.1 as DBA-DAMS-RE-1 to DBA-DAMS-RE-4, on the author's decision of 12 October 2026, and their instruments are owed by Project 11, which owns every reference-event instrument of the corpus. The use cases for the governing scenario of Section 6.3, for the cyber scenario of Section 6.4 and for the facilities in a dam's flood path have issued as DBA-DAMS-UC-1 to DBA-DAMS-UC-3, and the gaps they found for this class are taken in at this revision (Section 9.2). The register rows and the impacted sectors the instruments will propose are Project 0's to enter.

10.8 The Security Requirement for Dams Outside Federal Jurisdiction

The Commission's security program covers the dams it licenses and is guidance, and the Dams Sector guidance of the Cybersecurity and Infrastructure Security Agency is voluntary (Section 8.4). This instrument has found no federal security requirement for a dam outside the

Commission's jurisdiction, which includes most state-regulated dams. The gap is recorded, and its status is to be confirmed at each revision of this instrument.

10.9 What This Class Notes to Other Instruments

Several elements of this instrument belong to other instruments, and this instrument records them as noted without stating them for those instruments. To the framework: the reading of its gate-opening statement by the consequence of each command (Section 7.3), for the framework to confirm or replace; the event categories of Section 5, the inundation model of Section 4, the hazard information finding of Section 6.5, and the conditionality principle of Section 6.9; the conferral section it also asked for is now framework §9.1. To *Large Hydroelectric Storage*, in Project 8: the boundary of Section 10.3, now closed, and the condition on turbine discharge of Section 5.1, which that instrument keeps stating. To the nuclear series of DBA-MA, in Project 6: the Kakhovka reading of Section 6.3, that a nuclear plant on a reservoir carries the pool-dependency margin of Section 3.3, and that its own design basis should count the dam as a source of the loss of its cooling water. To the Emergency Services series (DBA-ESS) and the communications design basis: the requirement of Section 6.3 for a warning that works without the dam. And to the Chemical series (DBA-CHEM): that a plant in a dam's footprint counts the dam above it as a source of flood, the Midland plant having reported floodwater in its containment ponds in May 2020. From DBA-DAMS-UC-3 (*The Facilities in a Dam's Flood Path*), findings are noted to *Chemical Production and Synthesis* §3.3, that a plant within a dam's footprint takes as its design-basis flood the more severe of the mapped floodplain and the dam's deliberate-case footprint; to *Water Treatment Plant* §4.7, that a dam above the plant is a source of the exceedance; to *Response Forces and Stations* §5.4, that a positive applicability determination for a dam above an area is a dispersal trigger for the stations in its footprint; and to *Emergency Management and Coordination* §6.4, that the independence test applies to a measured-state trigger's detection and its link. Each is for that class to dispose of at its next revision. The two Emergency Services classes have since disposed of theirs on the author's decisions of 12 October 2026: *Response Forces and Stations* §5.4 and *Emergency Management and Coordination* §6.4 now state them.

10.10 Two Communications Design Basis Requirements Worded for the Electric Sector

CDB-2 and CDB-5 are read for the dam at Section 8.5 because their text names the electric sector. When the Military Action Communications Design Basis words them for every sector, Section 8.5 is revised to cite the new wording.

References

DBA-DAMS Series and Parent Framework

Dams Sector Framework (DBA-DAMS_Sector_Framework). Eclectic Technologies, 2026. Parent sector framework under which this instrument is issued.

DBA-MA Parent Framework. Eclectic Technologies. Source of the four military-action characteristics and the design-basis-under-military-action methodology.

Defense in Depth (DBA-MA-DiD). Eclectic Technologies, 2026. Independent-protection disposition step and declared-residual output form (§3.4, §3.5).

Military Action Communications Design Basis (DBA-MA-CDB). Eclectic Technologies, 2026. CDB-1 through CDB-7 (§2) and the instantiation requirement (§3).

Large Hydroelectric Storage (DBA-ES-HY-FC1). Eclectic Technologies, 2026. The electric series' hydroelectric class, whose dam-safety chain has moved to this class on the determination of 12 October 2026, and which cites this class for it (Section 10.3).

Storage Reservoir (DBA-W-FC3) and *Ways and Linehaul* (DBA-TRAN-WY-FC2). Eclectic Technologies, 2026. The water-supply and navigation boundaries.

Chemical Production and Synthesis (DBA-CHEM-PR-FC1), *Chemical Sector Framework* (DBA-CHEM_Sector_Framework) and *Petroleum Refining* (DBA-OIL-FC1). Eclectic Technologies, 2026. The structural model for this instrument and the precedents it follows for the provisional model home, hazard information and clustering.

The TRITON Attack on a Safety Instrumented System, 2017 (DBA-CHEM-RE-1). Eclectic Technologies, 2026. Cited by the short title *TRITON Attack* and finding number.

DBA-DAMS-RE-1, DBA-DAMS-RE-2, DBA-DAMS-RE-3 and DBA-DAMS-RE-4, conferred at framework §9.1 on 12 October 2026. Their instruments are owed by Project 11.

A Deliberate Breach at High Pool with the Warning Denied (DBA-DAMS-UC-1), *Gates Opened at High Pool Behind Normal Readings* (DBA-DAMS-UC-2) and *The Facilities in a Dam's Flood Path* (DBA-DAMS-UC-3). Eclectic Technologies, 2026. The use cases under this class, whose findings this revision takes in.

Foundational Definitions and Command Broker Family

FD-BL, FD-BR, FD-LD, FD-LS, FD-EV, FD-BL-D1 and FD-BL-D2, cited by designation; version state in FD-REG (*Governing Stack Register*). CB-Framework_Cross_Sector (*Command Broker Framework*), cited by designation; version state in CB-REG (*Command Broker Family Register*).

Regulatory, Standards and Investigations

18 CFR Part 12, Safety of Water Power Projects and Project Works, Subparts C and D; Federal Energy Regulatory Commission, Order No. 880, 87 FR 1490, 11 January 2022. Federal Energy Regulatory Commission, *Security Program for Hydropower Projects*, Revision 3A.

Federal Emergency Management Agency. *Federal Guidelines for Dam Safety: Hazard Potential Classification System for Dams* (FEMA P-333); *Federal Guidelines for Dam Safety: Emergency Action Planning for Dams* (FEMA P-64); *Federal Guidelines for Inundation Mapping of Flood Risks Associated with Dam Incidents and Failures* (FEMA P-946); *Federal Guidelines for Dam Safety Risk Management* (FEMA P-1025); flood risk analysis and mapping guidance for dams (Guidance Document 68), May 2016; National Inventory of Dams.

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United States Department of Justice. Press release 16-348, “Seven Iranians Working for Islamic Revolutionary Guard Corps-Affiliated Entities Charged for Conducting Coordinated Campaign of Cyber Attacks Against U.S. Financial Sector”, 24 March 2016.

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Revision History

Version	Date	Author	Description
0.1 — DRAFT	October 2026	Tim Roxey	Initial issue. First facility-class instrument in the DBA-DAMS series, typed physical under the author’s determination of 30 September 2026, in the ten-section structure of DBA-CHEM-PR-FC1. Establishes the three reference structures on the author’s decision of 12 October 2026 (a high-hazard concrete gravity dam with hydropower and a gated spillway, a large embankment dam

Version	Date	Author	Description
			with a gated spillway serving water supply, and a cascade of embankment dams on one river); the two consequence chains (inundation and loss of the pool); seven class design parameters; the non-adversarial event envelope; the military-action extension with the deliberate breach as its governing scenario; the Autonomous Safe Impoundment Condition and its three declared residuals; the dual design basis stated in full; and the required inheritance map. Carries, as its provisional home on the author's decision of 12 October 2026, the shared inundation consequence model the framework requires (Section 4). Proposes four reference events for the framework to confer, on the author's decision of the same day: Kakhovka 2023, Oroville 2017, Edenville–Sanford 2020 and Bowman Avenue 2013, each checked against primary sources where they could be reached. Cites the local-sovereignty property of protective functions to FD-LS directly and states the endurance requirement locally (OI-16). Proposes the boundaries with the coordination class and states those with the electric, water and transportation series, the electric boundary held open. Quantitative parameters and durations are flagged as pending.
0.2 — DRAFT	October 2026	Tim Roxey	Conforms to the framework at v0.4 and takes in the eight gaps DBA-DAMS-UC-1, UC-2 and UC-3 returned. The four reference events proposed at v0.1 are stated as conferred, as DBA-DAMS-RE-1 (Kakhovka, 2023), DBA-DAMS-RE-2 (Oroville, 2017), DBA-DAMS-RE-3 (Edenville and Sanford, 2020) and DBA-DAMS-RE-4 (Bowman Avenue, 2013), at Sections 5.3, 5.5, 5.7, 6.3, 9.2 and 10.7. From the use cases: Section 4.5 counts the warning time from the loss together of the dam's detection and warning paths, or from a divergence of independent indication, under a positive applicability determination, and Section 6.3 warns on it; Sections 5.4 and 5.5 require the independent indication to be compared continuously with the commanding system's and the divergence alarmed; Section 5.4 limits the opening rate of every gate whose release can reach the endpoint, below a stated pool and river condition, by its machinery, on the author's decision of 12 October 2026, and Section 7.2 counts the limit among the protective functions; Section 5.5 requires the record of remote access paths to be verified at the controllers; Section 6.3 states the basis on which the drawdown authority weighs the dependents' margins; Section 6.8 extends the second residual to an overtopping under a falsified basis; Section 8.1 requires every facility in the deliberate-case footprint to receive its site figures and holds the dam's design basis unmet until it has, on the author's decision of the same day, and Section 6.5 points to it; Section 10.6 extends the proposed boundary to detection off the structure. Section 6.4 cites DBA-DAMS-UC-2, and Section 10.9 records the conferral section as made and the findings DBA-DAMS-UC-3 notes to four classes

Version	Date	Author	Description
			of other series. Section 10.4 records OI-32. References add the conferred designators and the use cases. The revision-history table is set to the house widths. The cover lineage stays as issued. No change to the class's scope, chains, parameters, event categories or placements; the v0.1 entry is left as a record.
0.3 — DRAFT	October 2026	Tim Roxey	Conforms to the return note Project 8 sent this project on 12 October 2026 on the boundary with *Large Hydroelectric Storage* (DBA-ES-HY-FC1). On the author's determination of 12 October 2026 that instrument's dam-safety chain moves to this class, its generating chain and the interaction of the two chains staying with it; Sections 1.2, 8.3 and 10.3 state the move as agreed and its execution as pending at that instrument's next revision, and Section 10.3 records that this instrument governs in the meantime and that the drifts Project 8 found are not carried here. On the author's rulings of the same day on the three items the move offered: Section 5.1 counts turbine discharge in release capacity only while the wicket gates are shown operable, never against deliberate attack or toward the Autonomous Safe Impoundment Condition, and Section 5.6 says so; Section 5.8 and Table 5-1 take in the earthquake followed by an inflow before the structure is shown stable, the interval left to the site; and a new Section 8.5 applies DBA-MA-CDB (*Military Action Communications Design Basis*) CDB-1 to CDB-7 to the dam, taking in the dam-safety communications obligations, with Section 8.2 pointing to it, Table 9-1 adding the row, and a new Section 10.10 recording CDB-2 and CDB-5 as still worded for the electric sector. Section 6.7 states how a drawdown is ordered when the named authority cannot be reached. Section 10.2 records the 72-hour figure Project 8 offers to the framework. Sections 10.4 and 10.9 and the references are conformed. The cover lineage stays as issued. No change to the class's chains, parameters or placements; earlier entries are left as a record.
0.4 — DRAFT	October 2026	Tim Roxey	Closes the boundary with *Large Hydroelectric Storage* (DBA-ES-HY-FC1), on the notice Project 8 sent this project on 12 October 2026 that the revision carrying the move of the dam-safety chain has issued. Every citation of that instrument into this class was read against this class's text and lands on the section that states what it cites; Sections 1.2, 8.3 and 10.3 record the move as carried out, Section 10.4 records that instrument as a citer of Section 4, and the reference is conformed. Three citations found gaps in this class, taken in on the author's decision of 12 October 2026: Section 5.1 and Table 5-1 name waves on the pool, from wind or a landslide into the reservoir, beyond the freeboard, and count a wave over an embankment's crest as overtopping; Section 5.4 and Table 5-1 add the loss of power to the instruments on the structure,

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			and require backup power or stated manual observation for the instruments that detect a potential failure mode; and Section 6.2 adds persistence through a long denial of inspection, with the interval after which the structure is no longer credited with its shown stability stated under a positive applicability determination. Section 10.9 records the boundary as closed and the two Emergency Services findings as disposed of. The cover lineage stays as issued. No change to the class's chains, parameters or placements, and no section renumbered; earlier entries are left as a record.